

A Comprehensive Framework for Hallucination Detection in Large Language Models for Indian Languages

Towards Reliable and Trustworthy Multilingual AI

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The Hallucination Problem in Multilingual LLMs

Large Language Models such as LLaMA, GPT, and Gemini frequently generate **hallucinations** — factually incorrect, fabricated, or unsupported information presented with high confidence. This problem is significantly more severe in Indian regional languages due to script diversity, translation ambiguity, and the absence of dedicated multilingual benchmarks.

Key Challenges

- Most hallucination detection tools (SelfCheckGPT, FactCC, TruthfulQA) are English-only
- Indian languages — Hindi, Tamil, Kannada, Telugu, Bengali — are underrepresented in training data and evaluation benchmarks
- Traditional string-matching evaluation incorrectly classifies correct native-script responses as hallucinations
- **Example:** Model answers "नई दिल्ली" (correct) for "New Delhi" → flagged as HALLUCINATED by naive evaluation

Three Contributions of This Work

1. **MHDF** — Multilingual Hallucination Detection Framework
2. **ILHB** — Indian Language Hallucination Benchmark (150 QA pairs)
3. **Translation-Aware Hybrid Classifier** — the first of its kind for Indian language LLM evaluation



Research Gap: No existing framework comprehensively evaluates and explains hallucinations across multiple Indian languages with a unified, reproducible methodology.

Introduction & Research Scope

As LLMs expand into Indian-language deployments across education, healthcare, and government services, ensuring factual reliability becomes critical. Hallucination in regional languages is both more frequent and harder to detect than in English.

Languages Covered (5)

Hindi · Tamil · Kannada · Telugu · Bengali

Models Evaluated (2)

- LLaMA 3.1 8B Instant (via Groq API — free tier)
- LLaMA 3.3 70B Versatile (via Groq API — free tier)

Hallucination Types Studied

Factual · Entity · Numeric · Translation-Induced · Contextual

Knowledge Domains (6)

History · Geography · Science · Constitution · Culture · Personalities

Out of Scope

- Model fine-tuning or retraining
- Code-mixed or dialect input
- Languages beyond the five selected
- GPU or paid compute — entire pipeline runs free on Google Colab

Research Objectives

1**Study Hallucination Behaviour**

Examine hallucination patterns in LLMs across multiple Indian languages under controlled, reproducible conditions.

2**Translation-Aware Framework**

Develop a framework that correctly handles diverse scripts and language representations, resolving native-script equivalence mismatches.

3**Construct ILHB Benchmark**

Build the Indian Language Hallucination Benchmark — 150 QA pairs across 5 languages and 6 knowledge domains for systematic factual evaluation.

4**Compare Model Performance**

Compare factual accuracy of LLaMA 3.1 (8B) vs LLaMA 3.3 (70B) using a unified, translation-aware evaluation methodology.

5**Analyse Hallucination Patterns**

Perform language-wise and domain-wise analysis of hallucination patterns across multilingual AI systems.

Literature Survey — Comparative Summary

Paper	Method	Key Contribution	Limitation	Relevance to This Work
Ji et al. (2023)	Hallucination taxonomy	Formal types and metrics	English-centric	Foundation for hallucination typology
SelfCheckGPT (Manakul et al., 2023)	Self-consistency	Black-box hallucination detection	Computationally expensive; limited multilingual support	Baseline comparison
FactCC (Krycinski et al., 2020)	NLI-based verification	Source-grounded fact checking	English-only NLI models	Motivation for multilingual gap
TruthfulQA (Lin et al., 2021)	Benchmark	Truthfulness evaluation	No Indian language variants	Motivated ILHB construction
RAG (Lewis et al., 2020)	Retrieval-augmented generation	Evidence-supported grounding	Dependent on knowledge source coverage	Basis for Wikipedia retrieval in MHDF
IndicBERT (Kakwani et al., 2020)	Multilingual embeddings	Strong Indian language support	Not designed for hallucination detection	Informs multilingual representation
MHDF + ILHB (This work)	Translation-aware hybrid pipeline	First unified Indian-language hallucination framework	150 QA pairs; 5 languages; Wikipedia only	This study



Research Gap: No existing method provides a unified, translation-aware, reproducible hallucination detection framework covering all five target Indian languages simultaneously.

MHDF Pipeline — Proposed Methodology

01	02	03
ILHB Dataset 150 QA pairs · 5 languages · 6 domains	LLM Response Generation LLaMA 3.1 8B + LLaMA 3.3 70B via Groq API · Temperature 0.1 · Identical prompting conditions	Wikipedia Knowledge Retrieval Ground-truth answer used as search term · Retrieved summary = reference evidence (E)
04	05	06
Translation-Aware Verification (Key Contribution) Native-script responses translated to English via Google Translator API before comparison · Resolves: "नई दिल्ली" ≡ "New Delhi"	LaBSE Semantic Similarity Cosine similarity between response and evidence vectors · Score range: 0 to 1	Hybrid Hallucination Classifier (4 Stages) Stage 1: Direct string match → FACTUAL (1.0) · Stage 2: Numeric match → FACTUAL (0.95) · Stage 3: Translation match → FACTUAL (0.90) · Stage 4: Similarity threshold ≥ 0.70 → FACTUAL · 0.50–0.70 → UNCERTAIN · < 0.50 → HALLUCINATED · No evidence → UNVERIFIABLE
07		
Result Analysis Language-wise · Domain-wise · Model-wise · Method-wise		
Methodology Evolution: v1 (similarity only): 72.0% / 81.3% hallucination → implausibly high · v2 (+ string match): 53.3% / 62.0% → still inflated · Final (+ translation + manual): 21.1% / 14.4% → accurate		

ILHB — Indian Language Hallucination Benchmark

150

Total Questions

Manually verified QA pairs

5

Languages

Hindi, Tamil, Kannada, Telugu, Bengali

30

Questions per Language

Balanced distribution across all languages

6

Knowledge Domains

5 questions per domain per language

Domain	Sample Question	Correct Answer
History	India's first Prime Minister?	Jawaharlal Nehru
Geography	Capital of India?	New Delhi
Science	Chemical formula of water?	H ₂ O
Constitution	When did Indian Constitution come into effect?	26 January 1950
Culture	Which state celebrates Pongal?	Tamil Nadu
Personalities	Nobel Prize year of Rabindranath Tagore?	1913



All 150 QA pairs manually verified using Wikipedia, NCERT, and official government sources. No code-mixed or dialect inputs — standard written form only.

Requirement Specification

Software Stack

- Python 3.10+ | Google Colab (execution environment)
- [Groq API](#) — LLaMA 3.1 8B + LLaMA 3.3 70B (free tier)
- wikipedia-api — knowledge retrieval
- sentence-transformers — LaBSE embeddings
- deep-translator — Google Translator API
- Pandas, NumPy, Matplotlib — analysis and visualisation

Hardware & Infrastructure

- No GPU required
- All computation via free-tier cloud APIs
- Google Colab CPU runtime (free)
- Total infrastructure cost: ₹0

Dataset & Runtime

- 150 questions
- ~300 API calls total
- <15 min end-to-end runtime

Experimental Results — Translation-Aware Hybrid Classifier

Table 1 — Overall Classification Results

Model	Factual	Hallucinated	Uncertain	Unverifiable	Factual Accuracy	Hallucination Rate
LLaMA 3.1 (8B)	106	30	6	8	69.3%	21.1%
LLaMA 3.3 (70B)	124	21	1	4	83.3%	14.4%

Table 2 — Hallucination Rate by Language

Language	LLaMA 3.1 (8B)	LLaMA 3.3 (70B)
Hindi	7.1%	17.2%
Tamil	23.3%	16.7%
Kannada	28.6%	10.3%
Telugu	18.5%	10.7%
Bengali	31.0%	13.3%

Table 3 — Detection Method Distribution (LLaMA 3.1)

⋮

Method	Count
Translation Match	61 (41%)
Semantic Similarity	38 (25%)
Direct Match	34 (23%)
No Evidence	8 (5%)
Numeric Match	6 (4%)
Manual Review	3 (2%)

⋮

🔑 **Key Finding:** Translation-aware matching resolved 61 out of 150 responses — the single most influential detection mechanism. Without it, 72% of LLaMA 3.1 responses were mislabelled as hallucinated.

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